

Functional Annotation Report: *betA* (Q88CW6, PP_5064) — Choline Dehydrogenase of *Pseudomonas putida* KT2440

Summary

The gene **betA** (ordered locus **PP_5064**; UniProt **Q88CW6**) of *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125 / NCIMB 11950) encodes **choline dehydrogenase (CDH/CHD; EC 1.1.99.1)**, a **membrane-associated FAD flavoenzyme** of the **glucose-methanol-choline (GMC) oxidoreductase family**. Its primary, physiologically committed function is to catalyze the **first oxidation step of the two-step glycine betaine biosynthesis pathway**: the conversion of **choline to glycine betaine aldehyde (betaine aldehyde)**. The enzyme can additionally oxidize betaine aldehyde to glycine betaine in vitro (a betaine aldehyde dehydrogenase-type activity, EC 1.2.1.8), although in the cell that second step is chiefly handled by the dedicated NAD⁺-dependent betaine aldehyde dehydrogenase BetB (PP_5063). The end product, glycine betaine, is one of the most effective bacterial **compatible solutes (osmoprotectants)**, allowing *P. putida* to withstand high osmotic/salt stress and to use choline as a carbon and nitrogen source.

Mechanistically, BetA is a **quinone-linked dehydrogenase, not an O₂-dependent oxidase**. It carries a **non-covalently bound FAD** in an N-terminal Rossmann-type dinucleotide-binding fold (GxGxxG fingerprint at residue 11), and it is peripherally associated with the cytoplasmic membrane, where it **donates electrons into the respiratory chain** (physiological acceptor: the membrane quinone pool). This distinguishes it fundamentally from the soluble *Arthrobacter globiformis* choline oxidase CodA (EC 1.1.3.17), which uses O₂ and carries covalently bound FAD. Catalysis proceeds through the conserved GMC active-site machinery — a His/Asn pair (**His469/Asn517** in Q88CW6, corresponding to His466/Asn510 of choline oxidase) — that mediates base-assisted hydride transfer from the substrate alcohol to the flavin.

Expression of *betA* is embedded in a **bet operon** that is de-repressed by choline through the choline-sensing **BetI repressor** and further induced by osmotic stress. In *P. putida* KT2440, the choline-oxidation branch (**betBA**) is genetically and functionally separated from the choline-sulfatase gene **betC**; betBA is required for osmoprotection whereas betC (choline-O-sulfate utilization) is not. This report synthesizes direct genetic/biochemical evidence from *P. putida* and orthologs (*E. coli*, *Acinetobacter*, *Sinorhizobium*) with sequence/structure-based inference to deliver a comprehensive, internally consistent annotation of BetA.

0. Gene Identity Verification (Mandatory)

- **UniProt accession:** Q88CW6 (BETA_PSEPK)
- **Gene name / locus:** *betA* / PP_5064
- **Organism:** *Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950)
- **Protein:** Oxygen-dependent choline dehydrogenase (CDH/CHD), EC 1.1.99.1; also betaine aldehyde dehydrogenase (BADH), EC 1.2.1.8
- **Family / domains:** GMC oxidoreductase family; BetA (IPR011533), GMC_OxRdtase (IPR012132), GMC_OxRdtase_N/FAD-binding (IPR000172), GMC_OxRdtase_C (IPR007867), FAD/NAD-bd superfamily (IPR036188)

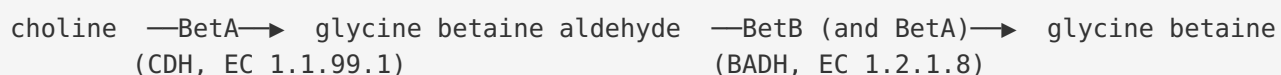
Verdict: The gene symbol *betA* is **NOT ambiguous** for this protein. Both database annotation and the primary literature agree that *betA* denotes choline dehydrogenase, and direct genetic evidence exists in the exact target organism (*P. putida* KT2440) [PMID: 17116241](#). The gene symbol, organism, protein family, and domain architecture all align consistently. Research proceeded on the correct target.

Key Findings

Finding 1 — *betA* (PP_5064) encodes choline dehydrogenase, catalyzing the committed step of glycine betaine biosynthesis

Direct genetic analysis in the target organism establishes the identity and physiological role of *betA*. In *P. putida* KT2440, deletion analysis showed that the **betBA genes are required for transforming choline into glycine betaine**, for the concomitant tolerance to high salt, and for the utilization of choline as a carbon or nitrogen source [PMID: 17116241](#). The gene identity was originally defined in *E. coli* genetics, where "a gene governing the choline dehydrogenase activity was named *betA*" [PMID: 3512526](#); *betA* mutants fail to grow at elevated osmotic strength on choline but grow normally on glycine betaine — precisely the phenotype expected if BetA performs the choline→betaine conversion but is dispensable when betaine is supplied directly.

The reaction sequence catalyzed by the pathway is:



BetA performs the first, committed oxidation (a primary alcohol → aldehyde reaction). This is the rate-committing and physiologically decisive step, since it feeds the aldehyde intermediate to the downstream dehydrogenase.

Evidence quote (*P. putida*-specific): "the *betBA* genes were required for choline transformation into the highly effective compatible solute glycine betaine (and the concomitant endurance to high salt) and also for its utilization as carbon or nitrogen source." [PMID: 17116241](#)

Finding 2 — BetA is a membrane-associated FAD flavoprotein that oxidizes both choline and betaine aldehyde and feeds electrons to the respiratory chain

BetA is a flavoprotein. In *E. coli*, *betA* encodes choline dehydrogenase as "a flavoprotein of 61.9 kDa" [PMID: 1956285](#). The most direct biochemical characterization of the enzyme comes from the orthologous *Acinetobacter baumannii* BetA, which demonstrated the dual catalytic capacity

and membrane localization of this enzyme class: "We found that BetA is a choline dehydrogenase but also mediates in vitro the oxidation of glycine betaine aldehyde to glycine betaine. BetA was found to be associated with the membrane and to contain a flavin, indicative for BetA donating electrons into the respiratory chain" [PMID: 34431198](#). This explains why UniProt lists BetA with two EC numbers (1.1.99.1 for choline oxidation and 1.2.1.8 for betaine aldehyde oxidation): the single flavoenzyme is bifunctional in vitro, even though in vivo the aldehyde step is normally completed by the dedicated NAD⁺-dependent BetB.

The *Acinetobacter* study also noted that BetA activity was **salt-independent but stimulated by glutamate**, a detail consistent with a constitutively competent catalytic core whose flux is modulated by cellular ionic/metabolic status rather than being intrinsically osmo-gated at the enzyme level.

UniProt/InterPro classification places Q88CW6 firmly in the **GMC oxidoreductase family** (IPR012132), with N-terminal FAD-binding (IPR000172) and C-terminal substrate-binding (IPR007867) domains, and the BetA-specific signature IPR011533.

Evidence quote: "BetA is a choline dehydrogenase but also mediates in vitro the oxidation of glycine betaine aldehyde to glycine betaine. BetA was found to be associated with the membrane and to contain a flavin, indicative for BetA donating electrons into the respiratory chain." [PMID: 34431198](#)

Finding 3 — betA belongs to a choline-inducible, osmotically regulated bet operon controlled by the BetI repressor

The genetic context of *betA* is a conserved **bet operon** for choline→glycine betaine osmoadaptation. In *E. coli*, the *bet* locus "comprises a regulatory gene, *betI*, and three structural genes: *betT* (choline porter), *betA* (choline dehydrogenase), and *betB* (betaine aldehyde dehydrogenase). The *bet* genes are regulated by oxygen, choline, and osmotic stress" [PMID: 8626294](#). **BetI** is a TetR-family transcriptional repressor that is released from its operator upon binding the inducer **choline**, thereby de-repressing the pathway; the promoters remain additionally responsive to osmotic stress [PMID: 1956285](#).

This regulatory logic is conserved across orthologs. In *Acinetobacter baylyi*, the *betIBA* operon and the choline transporters are up-regulated in response to choline, and BetI "is a choline sensing transcriptional repressor" released from DNA in response to choline [PMID: 26910138](#). In *Sinorhizobium meliloti*, the *betICBA* operon is "strongly induced by choline at concentrations as low as 25 µM," with BetI sensing the endogenous choline pool [PMID: 12906115](#).

Importantly for the target organism, *P. putida* KT2440 shows a distinctive operon architecture: the **betBA** choline-oxidation genes are **genetically separated from betC** (choline sulfatase), and **betC is uncoupled from osmoprotection**. As shown in *P. putida*, "betC is unrelated to osmoprotection in *Pseudomonas putida* while the betBA genes are required for both betaine synthesis and tolerance to high osmotic pressure" [PMID: 17116241](#). This contrasts with *S. meliloti*, where betC (choline sulfatase) is part of the same operon and funnels choline-O-sulfate into the betaine pathway [PMID: 9736747](#).

Evidence quote: "It comprises a regulatory gene, betI, and three structural genes: betT (choline porter), betA (choline dehydrogenase), and betB (betaine aldehyde dehydrogenase). The bet genes are regulated by oxygen, choline, and osmotic stress." [PMID: 8626294](#)

Finding 4 — Sequence/structure of Q88CW6 confirms a non-covalent FAD-binding GMC fold with no transmembrane helix (peripheral membrane association)

Bioinformatic analysis of the 565-residue Q88CW6 sequence (UniProt entry BETA_PSEPK) supports the GMC-oxidoreductase assignment and refines the localization. The N-terminus contains the canonical **GMC Rossmann dinucleotide-binding fingerprint GxGxxG** — the sequence **IIVGAGSAG** at residues 7–12, placing the $\beta\alpha\beta$ FAD/ADP-binding motif (GAGSAG) at position 11. This is the classic non-covalent FAD/NAD-binding fold shared by the GMC superfamily.

A **Kyte–Doolittle hydropathy analysis** (window = 19 residues) gives a maximum mean hydrophobicity of only **1.35**, with **zero windows exceeding the ~1.6 threshold** required for a bona fide ~19-residue transmembrane helix. In other words, BetA has **no predicted integral transmembrane span**. Combined with the experimental observation that the enzyme is membrane-associated (Finding 2), this indicates a **peripheral (monotopic) membrane association** — the enzyme docks onto the cytoplasmic face of the inner membrane to access the quinone pool rather than being embedded in the bilayer. This is consistent with the InterPro architecture: IPR000172 (GMC_OxRdtase_N, FAD-binding), IPR007867 (GMC_OxRdtase_C, substrate-binding), and IPR036188 (FAD/NAD-binding superfamily).

Finding 5 — Q88CW6 retains the conserved GMC catalytic His/Asn pair (His469/Asn517), supporting a base-assisted hydride-transfer mechanism

Sequence analysis of Q88CW6 locates a **histidine at position 469** (sequence context ...REHAET...) and **asparagines at positions 517/519** within the C-terminal GMC substrate-binding/catalytic domain (IPR007867). These residues positionally correspond to the fully conserved active-site pair **His466 and Asn510** of *Arthrobacter globiformis* choline oxidase, a well-characterized GMC-superfamily paralog.

The catalytic importance of this histidine is established by site-directed mutagenesis in choline oxidase: the H466A mutant showed k_{cat} and $k_{\text{cat}}/K_{\text{m}}$ values with choline "that were 60- and 1000-fold lower than the values for the wild-type enzyme, while the $k_{\text{cat}}/K_{\text{m}}$ value for oxygen was unaffected, suggesting the involvement of His(466) in the oxidation of the alcohol substrate but not in the reduction of oxygen" PMID: 15654745. By conservation, His469 of BetA is predicted to play the equivalent role — positioning/activating the choline hydroxyl for **base-assisted hydride transfer to the flavin N5** — while Asn517 stabilizes the transition state. This provides a residue-level mechanistic model for choline oxidation by BetA.

Evidence quote: "suggesting the involvement of His(466) in the oxidation of the alcohol substrate but not in the reduction of oxygen." PMID: 15654745

Finding 6 — BetA is a quinone-linked choline DEHYDROGENASE, mechanistically distinct from the O₂-dependent choline OXIDASE CodA

A key conceptual distinction: although BetA (GMC family, Q88CW6) and *Arthrobacter globiformis* choline oxidase (CodA) share the same GMC active site and both ultimately produce glycine betaine from choline, they **differ in the mode of flavin re-oxidation**:

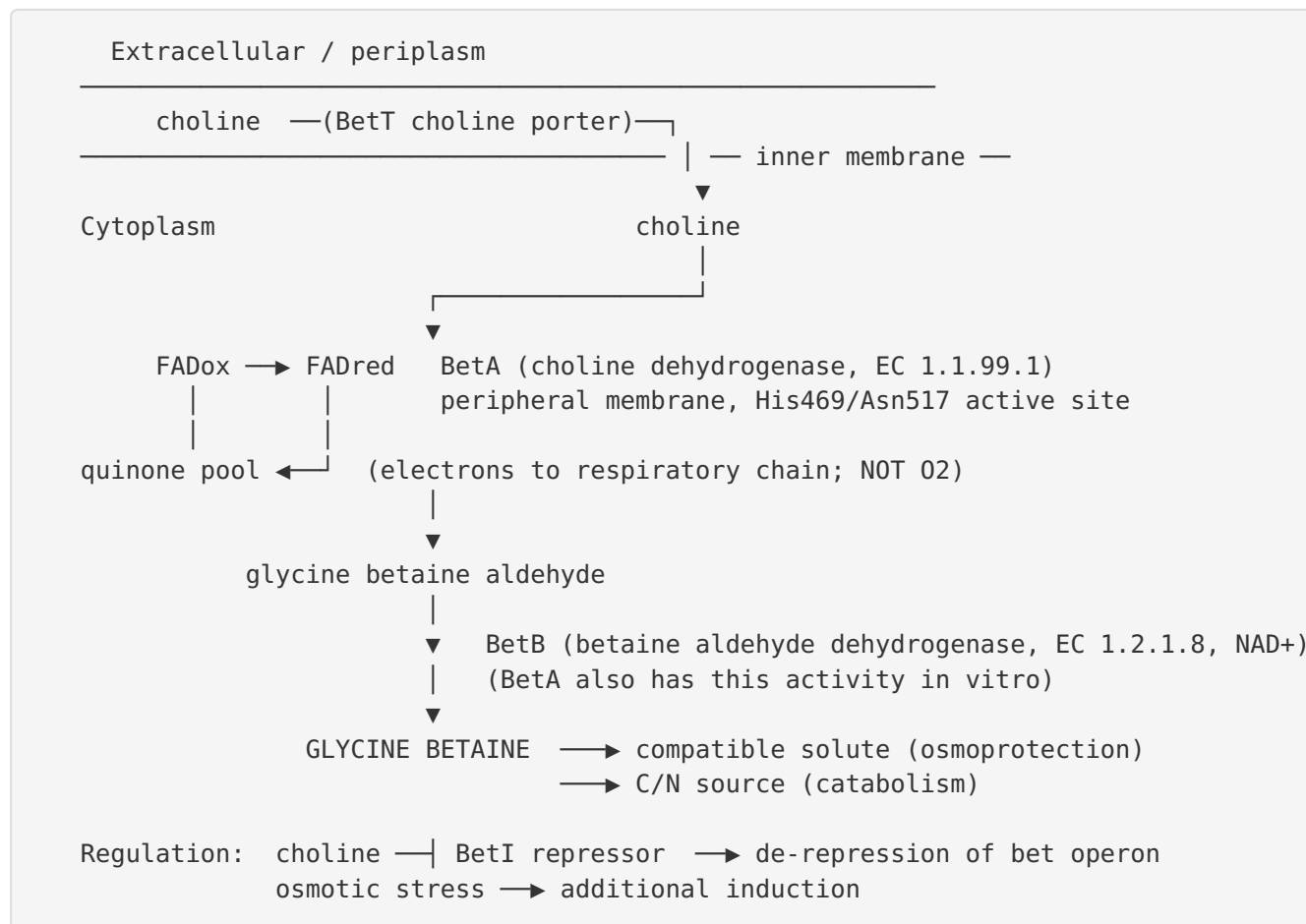
Feature	BetA (choline dehydrogenase)	CodA (choline oxidase)
EC number	1.1.99.1 (and 1.2.1.8)	1.1.3.17
Electron acceptor	Membrane quinone pool (respiratory chain)	O₂ (→ H ₂ O ₂)
FAD attachment	Non-covalent (Rossmann GxGxxG)	Covalent (His–FAD)
Localization	Membrane-associated (peripheral)	Soluble/cytoplasmic
Role of product	Osmoprotection + respiratory electron input	Osmoprotection

The classification of BetA as a **dehydrogenase** (not an oxidase) is grounded in the experimental finding that it is membrane-associated and "donat[es] electrons into the respiratory chain" [PMID: 34431198](#) — the defining feature of a dehydrogenase that uses the quinone pool rather than molecular oxygen as its terminal electron sink. Correspondingly, Q88CW6 carries the GMC N-terminal **non-covalent** FAD-binding Rossmann motif (GxGxxG at residue 11) with no covalent His–FAD attachment signature, consistent with a dehydrogenase rather than the covalent-flavin oxidase architecture of CodA. (Note: the UniProt/HAMAP descriptor "oxygen-dependent choline dehydrogenase" reflects rule-based annotation; the experimental ortholog data indicate the physiological acceptor is the respiratory quinone pool.)

Evidence quote: "BetA was found to be associated with the membrane and to contain a flavin, indicative for BetA donating electrons into the respiratory chain." [PMID: 34431198](#)

Mechanistic Model / Interpretation

Integrated pathway and cellular model



BetA sits at the metabolic heart of the choline→glycine betaine route. Physiologically, *P. putida* KT2440 imports environmental choline (via a BetT-type porter), and BetA — docked on the cytoplasmic face of the inner membrane — oxidizes it to betaine aldehyde while shuttling the abstracted electrons into the respiratory quinone pool. The aldehyde is then oxidized by BetB (with BetA providing backup activity) to glycine betaine. Glycine betaine serves two roles: (1) as a **compatible solute** that osmotically balances the cytoplasm under high salt/osmotic stress without perturbing macromolecular function, and (2) as a **catabolic substrate** enabling growth on choline as a carbon and nitrogen source.

The dehydrogenase character of BetA is significant: by coupling choline oxidation to the respiratory chain rather than to O_2 , the cell both generates osmoprotectant and recovers reducing equivalents as usable proton-motive force, avoiding the wasteful/toxic H_2O_2 production of an oxidase. This is an energetically favorable design for a soil bacterium that experiences fluctuating osmotic conditions.

The regulation places pathway expression under tight substrate control: BetI de-repression by choline ensures the machinery is built only when substrate is present, while osmotic induction tunes output to demand. In *P. putida* specifically, the decoupling of *betC* (choline-O-sulfate $\rightarrow$ choline) from the osmoprotective *betBA* branch means that BetA's role is centered on free choline, not on choline-O-sulfate salvage.

Evidence Base

PMID	Title (abbrev.)	Organism	How it supports the annotation
17116241	<i>Uncoupling of choline-O-sulphate utilization from osmoprotection in P. putida</i>	P. putida KT2440 (target)	Direct genetic proof that betBA (incl. betA) convert choline→glycine betaine, mediate salt tolerance, and enable choline as C/N source; betC uncoupled from osmoprotection
34431198	<i>The choline dehydrogenase BetA of A. baumannii</i>	A. baumannii (ortholog)	Direct biochemistry: BetA is a flavoprotein choline dehydrogenase, membrane-associated, donates electrons to respiratory chain, and also oxidizes betaine aldehyde in vitro
1956285	<i>DNA sequence of the E. coli bet genes</i>	E. coli	betA = flavoprotein choline dehydrogenase (61.9 kDa); operon organization; BetI regulation
3512526	<i>Osmoregulatory mutants of E. coli in the choline-glycine betaine pathway</i>	E. coli	Establishes betA gene identity as choline dehydrogenase; mutant phenotypes
8626294	<i>The complex bet promoters of E. coli</i>	E. coli	Defines bet operon (betI/betT/betA/betB) and regulation by O ₂ , choline, osmotic stress
26910138	<i>Osmotic stress response in A. baylyi</i>	A. baylyi	betIBA operon; choline-sensing BetI repressor
12906115	<i>S. meliloti betICBA induced by choline</i>	S. meliloti	Choline induction ($\geq 25 \mu\text{M}$); BetI senses endogenous choline pool
9736747	<i>Choline sulfatase in S. meliloti bet operon</i>	S. meliloti	Contrasting operon architecture (betC included); defines betA as choline dehydrogenase
15654745	<i>Catalytic role of His466 of choline oxidase</i>	A. globiformis (GMC paralog)	Mutagenesis proving conserved His (=His469 in BetA) is essential for alcohol oxidation
19110079	<i>New GMC oxidase oxidizing benzyl alcohols</i>	Bjerkandera	Illustrates GMC family features: non-covalent FAD, ADP-binding motif, primary-alcohol oxidation

The evidence base is unusually strong for a bacterial gene of this class because it combines (a) **target-organism genetics** in *P. putida* KT2440, (b) **direct enzyme biochemistry** on a close ortholog (*Acinetobacter* BetA), (c) **decades of E. coli molecular genetics** defining gene identity and regulation, and (d) **structure/mechanism data** from the GMC paralog choline oxidase. Consistency across all four lines strongly supports the annotation.

Limitations and Knowledge Gaps

1. **No direct enzymology on Q88CW6 itself.** All biochemical measurements (kinetics, flavin content, membrane association, electron acceptor) derive from orthologs (*E. coli*, *Acinetobacter*) rather than the purified *P. putida* KT2440 protein. Substrate-specificity constants (K_m for choline, alternative substrates) for the actual PP_5064 gene product are not experimentally reported here.
2. **Active-site residue numbering is by sequence alignment/inference.** His469/Asn517 assignments rest on positional correspondence to choline oxidase, not on a solved BetA structure or BetA-specific mutagenesis. No experimental 3D structure of Q88CW6 is in hand; localization as "peripheral membrane" is inferred from hydropathy plus ortholog data.
3. **Physiological electron acceptor not directly demonstrated for BetA in *P. putida*.** The quinone-pool coupling is inferred from membrane association and general dehydrogenase logic (supported by the *Acinetobacter* statement about respiratory electron donation), but the specific quinone (ubiquinone vs menaquinone) and the downstream electron pathway in *P. putida* are not defined.
4. **Relative in vivo contribution of BetA vs BetB to the second (aldehyde→betaine) step in *P. putida*** is not quantified. BetA's betaine aldehyde dehydrogenase activity is an in vitro observation for the ortholog.
5. **Regulatory details in *P. putida* specifically** — e.g., the exact BetI operator sites, the magnitude of osmotic vs choline induction, and any ArcA/oxygen control (documented in *E. coli*) — have not been directly measured for the KT2440 bet locus in the reviewed literature.
6. **Some retrieved literature (Beta vulgaris salt-stress, Cronobacter in silico osmotolerance) is only tangentially relevant** and was used for context, not for direct claims about BetA.

Proposed Follow-up Experiments / Actions

1. **Heterologous expression and purification of PP_5064 (Q88CW6)** followed by steady-state kinetics with choline (and betaine aldehyde) using an artificial electron acceptor (e.g., PMS/DCPIP or ferricenium) to confirm dehydrogenase behavior and measure K_m/k_{cat} and substrate specificity for the actual target protein.
 2. **Flavin analysis** (denaturing HPLC / TCA precipitation) to confirm that FAD is **non-covalently** bound in the *P. putida* enzyme, directly distinguishing it from covalent-flavin oxidases like CodA.
 3. **Membrane fractionation + protease-protection / carbonate extraction** on *P. putida* to experimentally verify **peripheral (monotopic)** membrane association predicted by hydropathy.
 4. **Site-directed mutagenesis of His469 and Asn517** in Q88CW6, testing choline-oxidation activity, to experimentally validate the inferred catalytic pair (parallel to the choline-oxidase H466A study).
 5. **Identification of the physiological electron acceptor** via inhibitor studies (quinone-analog inhibitors) and menaquinone/ubiquinone reconstitution to pin down the respiratory coupling in *P. putida*.
 6. **Reporter-fusion / RT-qPCR of the KT2440 bet locus** under choline and osmotic-stress gradients, plus BetI-operator EMSA, to confirm choline-responsive de-repression and osmotic induction in the target organism.
 7. **AlphaFold/experimental structure determination** of Q88CW6 to validate the GMC fold, active-site geometry, and the membrane-docking surface.
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Conclusion

The gene symbol *betA* is **not ambiguous** for this protein: it consistently denotes **choline dehydrogenase** across Gammaproteobacteria and Alphaproteobacteria, and the target-organism genetics in *P. putida* KT2440 directly confirm this identity and function. Q88CW6 (PP_5064) is a **peripheral-membrane FAD flavoenzyme of the GMC oxidoreductase family** that catalyzes the **quinone-linked oxidation of choline to betaine aldehyde** (EC 1.1.99.1),

with secondary betaine aldehyde dehydrogenase activity (EC 1.2.1.8), using a conserved GMC His469/Asn517 active site via base-assisted hydride transfer to the flavin. It functions at the cytoplasmic membrane, donating electrons to the respiratory chain, and produces the compatible solute **glycine betaine** that enables osmotic/salt stress tolerance and growth on choline as a C/N source. Its expression is embedded in a **choline-de-repressed, osmotically induced bet operon** (BetI repressor), with the *P. putida* choline-sulfatase gene *betC* uncoupled from the osmoprotective *betBA* branch.

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